GENERAL CHEMISTRY 2

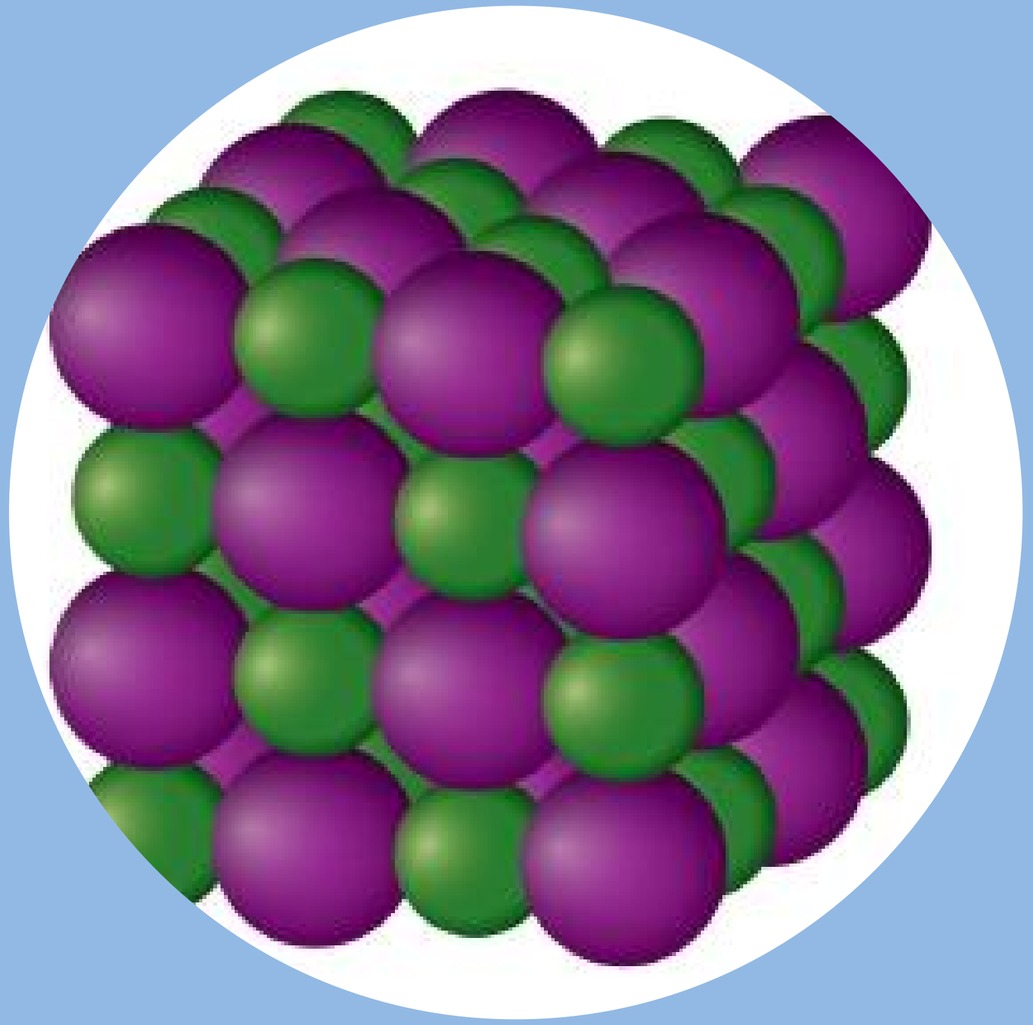


LESSON 4

TYPES AND PROPERTIES OF SOLIDS



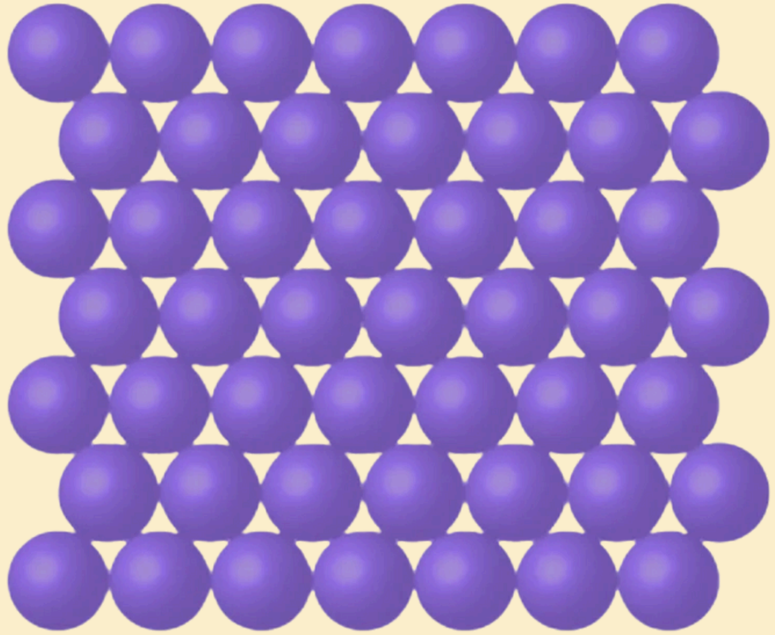
How do you describe solids?



TYPES AND PROPERTIES OF SOLIDS

CRYSTALLINE SOLIDS

have highly regular arrangement of particles.

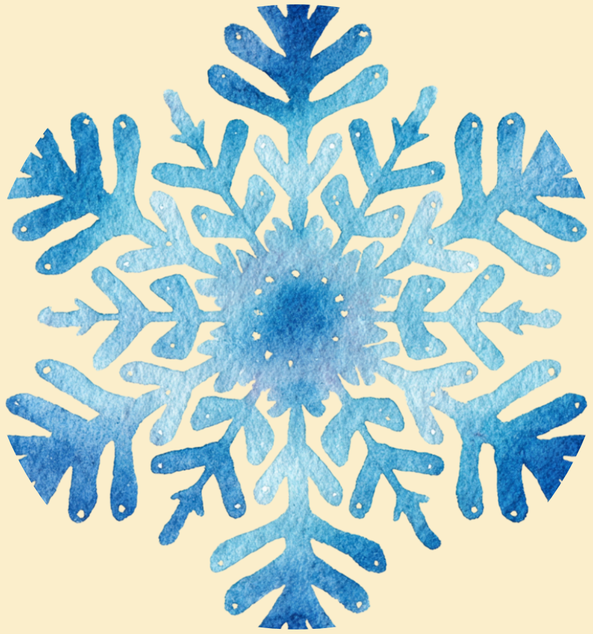


TYPES AND PROPERTIES OF SOLIDS

EXAMPLES



quartz



snowflakes



diamond

TYPES AND PROPERTIES OF SOLIDS

EXAMPLES



rock salt



rock

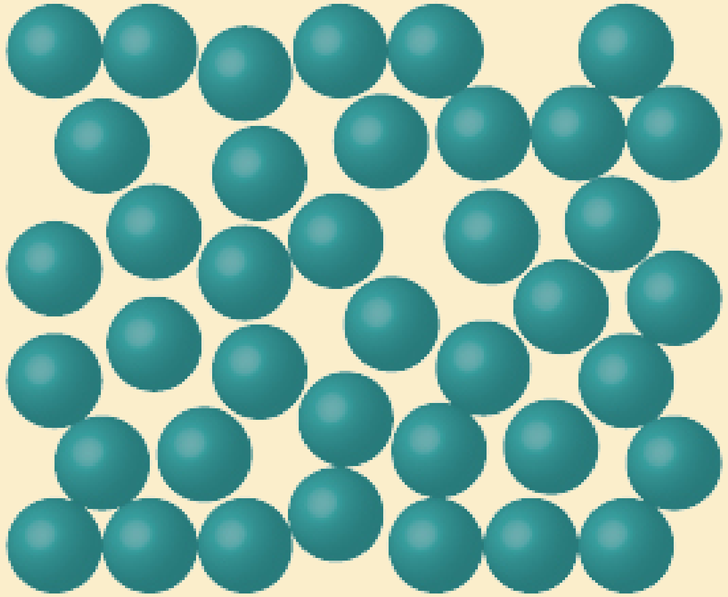


sugar

TYPES AND PROPERTIES OF SOLIDS

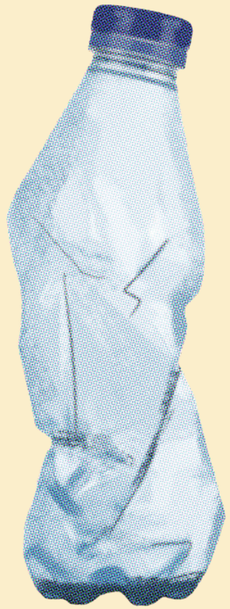
AMORPHOUS SOLIDS

any non-crystalline solid that does not organize the atoms and molecules in a definite lattice pattern.



TYPES AND PROPERTIES OF SOLIDS

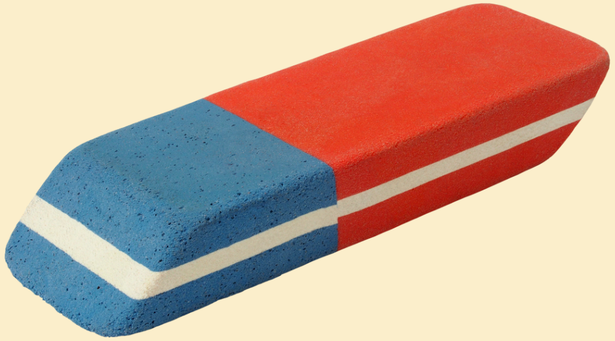
EXAMPLES



plastic



glass



rubber

TYPES AND PROPERTIES OF SOLIDS

EXAMPLES



gel



wax



fused silica

LESSON 4

PHASE CHANGES AND PHASE DIAGRAM

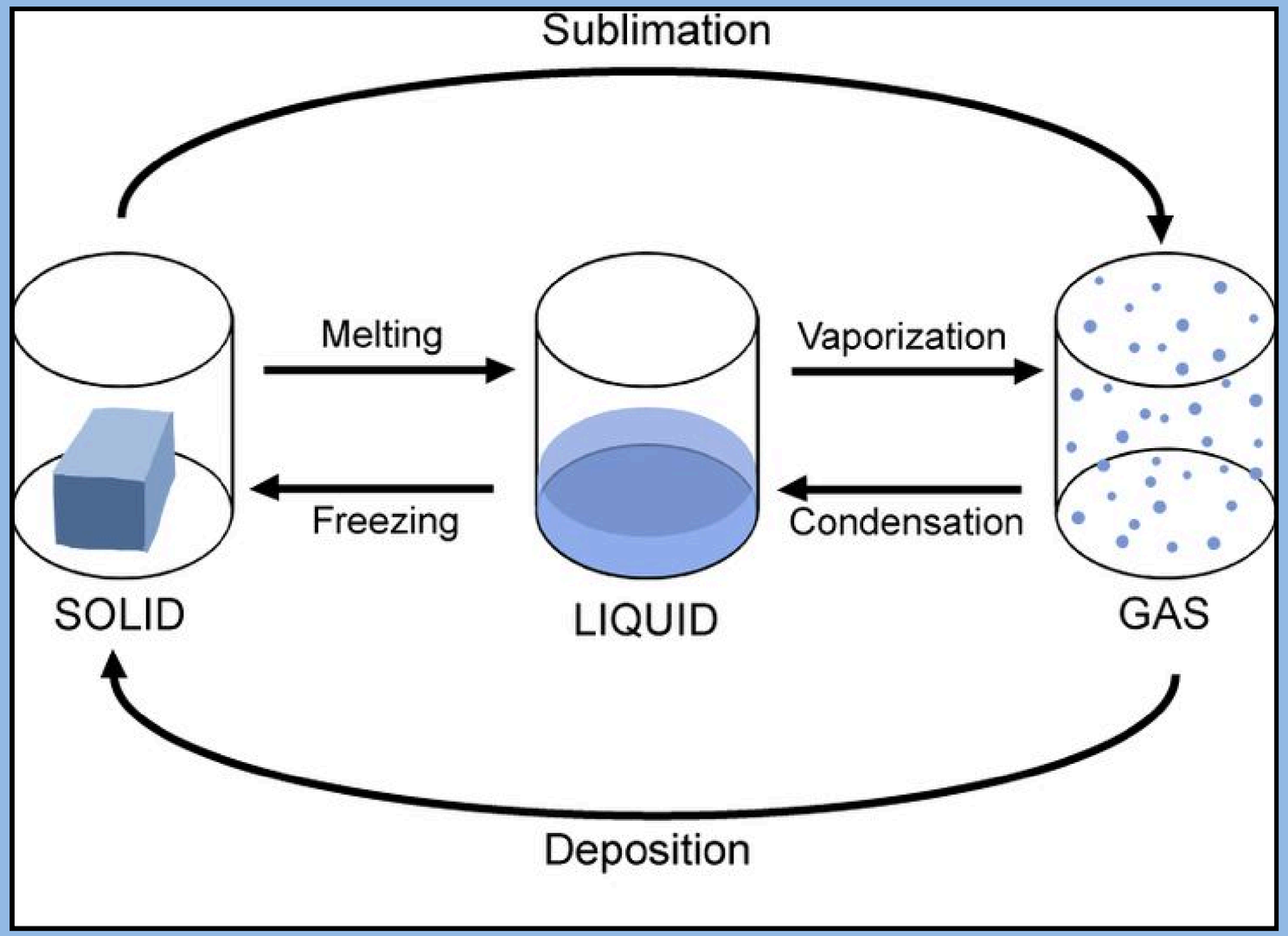


PHASE CHANGE

Transformation of matter from one physical state to another.

They occur when energy is added or removed from a substance.

TYPES OF PHASE CHANGE



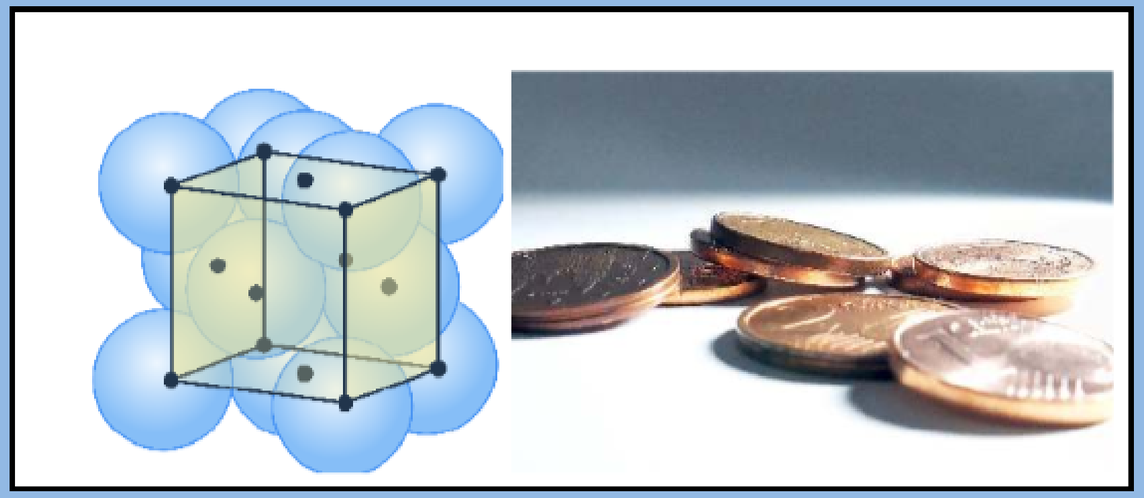
LESSON 5

TYPES AND PROPERTIES OF CRYSTALLINE SOLIDS



METALLIC CRYSTALS

Simply known as metals, consist entirely of metal atoms. They have string bonding due to dispersion forces. They also tend to adopt structures with symmetric, closed-packed arrangement of atoms.



IONIC CRYSTALS

Ionic crystals consist of cations and anions arranged in a lattice, held together by strong electrostatic attractions. They have high melting/boiling points, are hard but brittle, usually dissolve in water, and conduct electricity only when molten or dissolved (not as solids).



sodium chloride (NaCl)

magnesium oxide (MgO)



MOLECULAR CRYSTALS

Molecular crystals are made of neutral molecules held together by weak forces. They usually melt easily, are soft, and don't conduct electricity.



ice (H_2O)



dry ice (CO_2)



naphthalene (mothballs)

COVALENT NETWORK (GIANT COVALENT) CRYSTALS

Covalent network crystals are solids where all atoms are strongly linked together, like a giant “bonded web”. Because of this, they are very hard, have very high melting points, and usually don’t dissolve or conduct electricity



diamond



quartz

ACTIVITY 3

Direction: Fill in each box with the appropriate property for each type of crystalline solid.

Property	Ionic	Molecular	Covalent Network	Metallic
1. Melting point				
2. Hardness				
3. Brittleness				
4. Solubility in water				